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Differential Equations

GATE ECE Engineering Mathematics · Previous Year Questions with Answer Key

34	26	1994 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Answer key	at the end

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Marks 1 18 questions

Q1. GATE ECE 2026

Open solution

Consider the differential equation $\dot{\vec{w}} = A\vec{w}$, with $\vec{w}(t=0) = \begin{bmatrix}1\\1\end{bmatrix}$.

If $\vec{w}(t) = e^{t}\vec{u}_x + e^{-2t}\vec{u}_y$ be the solution to the equation where $\vec{u}_x$ and $\vec{u}_y$ are unit vectors along the positive x and y axes respectively, then which of the following options is the correct matrix representing A ?

(A) $\begin{bmatrix}1 & 0\\0 & -2\end{bmatrix}$

(B) $\begin{bmatrix}-1 & 0\\0 & 2\end{bmatrix}$

(C) $\begin{bmatrix}0 & -2\\1 & 0\end{bmatrix}$

(D) $\begin{bmatrix}0 & 2\\-1 & 0\end{bmatrix}$

Q2. GATE ECE 2025

Open solution

The function $y(t)$ satisfies

$t^2y''(t) - 2ty'(t) + 2y(t) = 0$

where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively. Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0, 1]$ is _____ (rounded off to two decimal places).

Q3. GATE ECE 2024

Open solution

The general form of the complementary function of a differential equation is given by $y(t) = (At + B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is ____.

(A) $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = f(t)$

(B) $\frac{d^2y}{dt^2} + 4y = f(t)$

(C) $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = f(t)$

(D) $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = f(t)$

Q4. GATE ECE 2022

Open solution

Consider the following partial differential equation (PDE)

$a\frac{\partial^2 f(x,y)}{\partial x^2} + b\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y),$

where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x,y) = e^{\xi x + \eta y}$ is a solution of the given PDE.

(A) $\xi = \frac{1}{\sqrt{2a}}, \eta\frac{1}{\sqrt{2b}}$

(B) $\xi = \frac{1}{\sqrt{a}}, \eta = 0$

(C) $\xi = 0,\ \eta = 0$

(D) $\xi = \frac{1}{\sqrt{a}}, \eta\frac{1}{\sqrt{b}}$

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Q5. GATE ECE 2021

Consider the differential equation given below. $\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$

The integrating factor of the differential equation is

(A) $(1-x^2)^{-\frac{1}{2}}$ (B) $(1-x^2)^{-\frac{3}{4}}$ (C) $(1-x^2)^{-\frac{3}{2}}$ (D) $(1-x^2)^{-\frac{1}{4}}$

Open solution

Q6. GATE ECE 2020

The general solution of $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$ is

(A) $y = C_1e^{3x} + C_2e^{-3x}$

(B) $y = C_1e^{3x}$

(C) $y = (C_1 + C_2x)e^{3x}$

(D) $y = (C_1 + C_2x)e^{-3x}$

Open solution

Q7. GATE ECE 2017 Set 2

The general solution of the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 5y = 0$ in terms of arbitrary constants K_1 and K_2 is

(A) $K_1e^{\left(-1+\sqrt{6}\right)x} + K_2e^{\left(-1-\sqrt{6}\right)x}$

(B) $K_1e^{\left(-1+\sqrt{8}\right)x} + K_2e^{\left(-1-\sqrt{8}\right)x}$

(C) $K_1e^{\left(-2+\sqrt{6}\right)x} + K_2e^{\left(-2-\sqrt{6}\right)x}$

(D) $K_1e^{\left(-2+\sqrt{8}\right)x} + K_2e^{\left(-2-\sqrt{8}\right)x}$

Open solution

Q8. GATE ECE 2017 Set 1

Consider the following statements about the linear dependence of the real valued functions $y_1 = 1,\ y_2 = x$ and $y_3 = x^2$. Over the field of real numbers.

I. y_1, y_2 and y_3 are linearly independent on $-1 \le x \le 0$

II. y_1, y_2 and y_3 are linearly dependent on $0 \le x \le 1$

III. y_1, y_2 and y_3 are linearly independent on $0 \le x \le 1$

IV. y_1, y_2 and y_3 are linearly dependent on $-1 \le x \le 0$

Which one among the following is correct?

(A) Both I and II are true

(B) Both I and III are true

(C) Both II and IV are true

(D) Both III and IV are true

Open solution

Q9. GATE ECE 2015 Set 2

The general solution of the differential equation $\frac{dy}{dx} = \frac{1+\cos 2y}{1-\cos 2x}$ is

(A) $\tan y - \cot x = C$

(B) $\tan x - \cot y = C$

(C) $\tan y + \cot x = C$

(D) $\tan x + \cot y = C$

Open solution

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Q10. GATE ECE 2015 Set 2

Consider the differential equation $\frac{dx}{dt} = 10 - 0.2\,x$ with initial condition $x(0) = 1$. The response $x(t)$ for $t > 0$ is

(A) $2 - e^{-0.2t}$ (B) $2 - e^{0.2t}$ (C) $50 - 49\,e^{-0.2t}$ (D) $50 - 49\,e^{0.2t}$

Open solution

Q11. GATE ECE 2014 Set 4

If a and b are constants, the most general solution of the differential equation $\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + x = 0$ is

(A) ae^{-t} (B) $ae^{-t} + bte^{-t}$ (C) $ae^{t} + bte^{-t}$ (D) ae^{-2t}

Open solution

Q12. GATE ECE 2014 Set 2

If the characteristic equation of the differential equation $\frac{d^2y}{dx^2} + 2\alpha\frac{dy}{dx} + y = 0$ has two equal roots, then the values of α are

(A) ± 1 (B) $0, 0$ (C) $\pm j$ (D) $\pm 1/2$

Open solution

Q13. GATE ECE 2012

With initial condition $x\,(1) \;=\; 0.5,$ the solution of the differential equation, $t\frac{dx}{dt} + x = t$ is

(A) $x = t - \frac{1}{2}$ (B) $x = t^2 - \frac{1}{2}$ (C) $xt = \frac{t^2}{2}$ (D) $x = \frac{t}{2}$

Open solution

Q14. GATE ECE 2011

The solution of differential equation $\frac{dy}{dx} = ky, y\,(0) = C$ is

(A) $x = Ce^{ky}$ (B) $x = ke^{Cy}$ (C) $y = e^{kx}C$ (D) $y = Ce^{-kx}$

Open solution

Q15. GATE ECE 2009

The order of differential equation $\frac{d^2y}{dt^2} + \left(\frac{dy}{dx}\right)^3 + y^4 = e^{-t}$ is

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q16. GATE ECE 2005

The following differential equation has $3\frac{d^2y}{dt^2} + 4\left(\frac{dy}{dt}\right)^3 + y^2 + 2 = x$

(A) degree $= 2$, order $= 1$ (B) degree $= 1$, order $= 2$

(C) degree $= 4$, order $= 3$ (D) degree $= 2$, order $= 3$

Open solution

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Q17. GATE ECE 2005

A solution of the differential equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$ is given by

(A) $y = e^{2x} + e^{-3x}$ (B) $y = e^{2x} + e^{3x}$ (C) $y = e^{-2x} + e^{3x}$ (D) $y = e^{-2x} + e^{-3x}$

Open solution

Q18. GATE ECE 1994

$y = e^{-2x}$ is a solution of the differential equation $y'' + y' - 2y = 0$

(A) TRUE (B) FALSE

Open solution

Marks 2 16 questions

Q19. GATE ECE 2020

Which one of the following options contains two solutions of the differential equation $\frac{dy}{dx} = (y-1)x$?

(A) $\ln|y-1| = 0.5x^2 + c$ and $y = -1$ (B) $\ln|y-1| = 2x^2 + c$ and $y = -1$

(C) $\ln|y-1| = 0.5x^2 + c$ and $y = 1$ (D) $\ln|y-1| = 2x^2 + c$ and $y = 1$

Open solution

Q20. GATE ECE 2018

The position of a particle y(t) is described by the differential equation :

$\frac{d^2y}{dt^2} = -\frac{dy}{dt} - \frac{5y}{4}$.

The initial conditions are y(0) = 1 and $\left.\frac{dy}{dt}\right|_{t=0}$ = 0.

The position (accurate to two decimal places) of the particle at t = π is _____.

Open solution

Q21. GATE ECE 2018

A curve passes through the point (x = 1, y = 0) and satisfies the differential equation

$\frac{dy}{dx} = \frac{x^2+y^2}{2y} + \frac{y}{x}$. The equation that describes the curve is

(A) $\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$ (B) $\frac{1}{2}\ln\left(1 + \frac{y^2}{x^2}\right) = x - 1$

(C) $\frac{1}{2}\ln\left(1 + \frac{y}{x}\right) = x - 1$ (D) $\ln\left(1 + \frac{y}{x}\right) = x - 1$

Open solution

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Q22. GATE ECE 2017 Set 1

Which one of the following is the general solution of the first order differential equation $\frac{dy}{dx} = (x + y - 1)^2$, where x, y are real ?

(A) $y = 1 + x + \tan^{-1}(x + c)$, where c is a constant

(B) $y = 1 + x + \tan(x + c)$, where c is a constant

(C) $y = 1 - x + \tan^{-1}(x + c)$, where c is a constant

(D) $y = 1 - x + \tan(x + c)$, where c is a constant

Open solution

Q23. GATE ECE 2016 Set 3

The particular solution of the initial value problem given below is $\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$ with $y(0) = 3$ and $\frac{dy}{dx}\Big|_{x=0} = -36$

(A) $(3 - 18x)e^{-6x}$

(B) $(3 + 25x)e^{-6x}$

(C) $(3 + 20x)e^{-6x}$

(D) $(3 - 12x)e^{-6x}$

Open solution

Q24. GATE ECE 2015 Set 3

Consider the differential equation $\frac{d^2x(t)}{dt^2} + 3\frac{dx(t)}{dt} + 2x(t) = 0$
Given $x(0) = 20$ & $x(1) = \frac{10}{e}$, where $e = 2.718$,
The value of $x(2)$ is

Open solution

Q25. GATE ECE 2015 Set 1

The Solution of the differential equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$ with $y(0) = y^1(0) = 1$ is

(A) $(2 - t)e^t$

(B) $(1 + 2t)e^{-t}$

(C) $(2 + t)e^{-t}$

(D) $(1 - 2t)e^t$

Open solution

Q26. GATE ECE 2014 Set 4

With initial values $y(0) = y'(0) = 1$, the solution of the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$ at $x = 1$ is _____.

Open solution

Q27. GATE ECE 2014 Set 3

Which ONE of the following is a linear non - homogeneous differential equation , where x and y are the independent and dependent variables respectively?

(A) $\frac{dy}{dx} + xy = e^{-x}$

(B) $\frac{dy}{dx} + xy = 0$

(C) $\frac{dy}{dx} + xy = e^{-y}$

(D) $\frac{dy}{dx} + e^{-y} = 0$

Open solution

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Q28. GATE ECE 2010

A function $n(x)$ satisfies the differential equation $\frac{d^2n(x)}{dx^2} - \frac{n(x)}{L^2} = 0$ where L is a constant. The boundary conditions are $n(0) = k$ and $n(\propto) = 0$. The solution to this equation is

(A) $n(x) = k \exp\left(\frac{-x}{L}\right)$

(B) $n(x) = k \exp\left(\frac{-x}{\sqrt{L}}\right)$

(C) $n(x) = k^2 \exp\left(\frac{-x}{L}\right)$

(D) $n(x) = k^2 \exp\left(\frac{-x}{\sqrt{L}}\right)$

Open solution

Q29. GATE ECE 2009

Match each differential equation in Group I to its family of solution curves from Group II.

Group I

$P: \frac{dy}{dx} = \frac{y}{x}$
$Q: \frac{dy}{dx} = \frac{-y}{x}$
$R: \frac{dy}{dx} = \frac{x}{y}$
$S: \frac{dy}{dx} = \frac{-x}{y}$

Group II

(1) Circle
(2) straight lines
(3) Hyperbola

(A) $P-2, Q-3, R-3, S-1$

(B) $P-1, Q-3, R-2, S-1$

(C) $P-2, Q-1, R-3, S-3$

(D) $P-3, Q-2, R-1, S-2$

Open solution

Q30. GATE ECE 2008

Which of the following is a solution to the differential equation $\frac{d}{dt}x(t) + 3x(t) = 0,\ x(0) = 2$?

(A) $x(t) = 3e^{-t}$

(B) $x(t) = 2\,e^{-3t}$

(C) $x(t) = \frac{-3}{2}t^2$

(D) $x(t) = 3t^2$

Open solution

Q31. GATE ECE 2007

The solution of the differential equation $k^2\frac{d^2y}{d\,x^2} = y - y_2$ under the boundary conditions (i) $y = y_1$ at $x = 0$ and (ii) $y = y_2$ at $x = \propto$ where k, y_1 and y_2 are constant is

(A) $y = (y_1 - y_2)e^{\frac{-x}{k^2}} + y_2$

(B) $y = (y_2 - y_1)e^{\frac{-x}{k}} + y_1$

(C) $y = (y_1 - y_2)\ \sin h\left(\frac{x}{k}\right) + y_1$

(D) $y = (y_1 - y_2)e^{\frac{-x}{k}} + y_2$

Open solution

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Q32. GATE ECE 2006

Open solution

For the differential equation $\frac{d^2y}{dx^2} + k^2y = 0$, the boundary conditions are
(i) $y = 0$ for $x = 0$ and
(ii) $y = 0$ for $x = a$
The form of non-zero solution of y (where m varies over all integrals) are

(A) $y = \sum_m A_m \sin\left(\frac{m\pi x}{a}\right)$
(B) $y = \sum_m A_m \cos\left(\frac{m\pi x}{a}\right)$
(C) $y = \sum_m A_m\ X^{\frac{m\pi x}{a}}$
(D) $y = \sum_m A_m\ e^{\frac{m\pi x}{a}}$

Q33. GATE ECE 2001

Open solution

Solve the differential equation $\frac{d^2y}{dx^2} + y = x$ with the following conditions (i) at $x = 0, y = 1$ (ii) at $x = 0, y' = 1$

Q34. GATE ECE 1994

Open solution

Match each of the items P, Q, R with an appropriate item from $1, 2, 3, 4$ and 5

List-I

$(P)\ a_1\frac{d^2y}{dx^2} + a_2y\frac{dy}{dx} + a_3y = a_4$
$(Q)\ a_1\frac{d^2y}{dx^3} + a_2y = a_3$
$(R)\ a_1\frac{d^2y}{dx^2} + a_2x\frac{dy}{dx} + a_3x^2y = 0$

List-II

(1) Non-linear differential equation
(2) Linear differential equation with constants coefficients
(3) Linear homogeneous differential equation
(4) Non-linear homogeneous differential equation
(5) Non-linear first order differential equation

(A) $P-1, Q-2, R-3$
(B) $P-3, Q-4, R-2$
(C) $P-2, Q-5, R-3$
(D) $P-3, Q-1, R-2$

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Answer Key

Marks 1

1. **A**	2. **0.25**	3. **A**	4. **A, B**	5. **D**
6. **C**	7. **A**	8. **B**	9. **C**	10. **C**
11. **B**	12. **A**	13. **D**	14. **C**	15. **B**
16. **B**	17. **B**	18. **A**		

Marks 2

19. **C**	20. **-0.23 to -0.19**	21. **A**	22. **D**	23. **A**
24. **0.85 to 0.86**	25. **B**	26. **0**	27. **A**	28. **A**
29. **A**	30. **B**	31. **D**	32. **A**	33. $y = x + \cos x$
34. **A**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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